

Course milestone M6: Data and Measurement Governance

HONR 46400 · Evidence-Driven Research

Start here

Your main file is `lastname_m06.pdf`. Build it in this order:

1. **Your carry-forward record, at the very top.** Open my review email and your research action plan side by side. Answer every numbered request for this milestone (each has an ID like `IR-M06-01`) and every step your plan sets for this milestone. Next to each one, write **Done** and exactly where I can find it, or **Not adopted**, with your reason and what you chose instead. Several items can point to the same piece of work.
2. **Then your milestone work and files.** The book milestone below explains the research work. The complete course checklist, the files to hand in and the rubric are in this milestone's assignment on Brightspace: check them before you submit.
3. **The Expo work** in *What this course adds*, below.

How this counts. Your answer to the action plan is worth 10 points of the rubric, scored for you individually. Each numbered request left unanswered costs 5 points, up to 20. Only requests and plan steps I sent you at least 48 hours before the deadline count; anything later moves to the next milestone. A step that is also a numbered request is counted once, as a request. If I sent you no numbered requests, or no plan steps apply, say so in one line. Not sure what applies to you? Write to me before the deadline; that is exactly what I am here for.

Submit it on Brightspace, under **Assignments** then **M06**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 6: Your data and measurement, governed** (checkpoint, version 1) of the course book, EDR|AI.

What this course adds

Your project title and the first draft of your conference abstract

You apply to present at the Undergraduate Research Expo at the next milestone, and your title and abstract go out with that application. An abstract written the night of the application has had no reader, so the title and the first draft come in now, and the book milestone does not ask for them.

Follow the conference's limits from the start: the submission portal caps the title at 125 characters and the abstract at 2,000 characters, spaces included in both. Count characters, not words, and capitalize the first letter of each major word in the title.

Your title highlights your research problem and the answer you are working toward. A reader who sees only the title should know what you are studying and what kind of answer the project will deliver.

Your abstract is one structured paragraph. Write it as a single paragraph, with no line breaks, that moves through four parts in this order. The order of the parts carries the structure.

1. **Question, answer, context, contribution: four sentences.** The first sentence is your research question. The second is your answer to it; you have no result yet, so state the answer you expect, plainly as an expectation, never as a finding. The third gives the background context, including the main conclusion of the literature on your question, then the research gap that conclusion leaves, your primary objective, and your main hypothesis: the testable expectation your design will confront. State the literature's conclusion from sources you have read and verified, never from an AI summary. The fourth states the contribution your project aims to make to the field: how your answer will add value to that main conclusion of the literature, for example by extending it, qualifying it, testing it in a new setting, or challenging it.

2. **Your data.** A description of the dataset you governed this week, or of what you expect it to be if it is not in hand yet: its source, units, coverage, and the measures you built. Where you can, add one or two results from your exploratory data analysis (EDA), the first look at your data's counts, ranges, and distributions, with the insight each one gives.

In the book: Lesson 20, Data Provenance and Data Quality, shows how to say where each value comes from and what it is a value of; its worked example traces a county turnout figure back to the clerk's certified canvass and finds three different turnout rates behind three denominators. Lesson 21, Measurement and Operationalization, shows how to describe the measures you built, from concept to construct to indicator. An EDA insight is exploratory, a pattern you noticed after seeing the data, so label it that way and never present it as a test (Lesson 12, Preregistration and the pre-analysis plan).

Example, built from the worked example in Lesson 21: "The data are 960 survey answers from employees at 16 work sites, 60 per site, eight that piloted flexible scheduling and eight that kept fixed shifts. Engagement is measured through one facet, willingness to recommend the workplace, as the mean answer to a single 1–7 agreement item; re-asking a random tenth of respondents two weeks later returned closely matching answers."

3. **Your study design.** The design, the sample size, the tools that collect the data, and the analytical technique you will use.

In the book: Lesson 10, Model, Inquiry, Data Strategy, and Answer Strategy, gives you the words for this part. Your data strategy is who is sampled, who gets which condition, and how each outcome is measured: your sample and your data-collection tools. Your answer strategy is the estimate you will compute and how you will state its uncertainty: your analytical technique. Lesson 12, Declaring and Diagnosing a Research Design, shows why a sample size is justified, not just reported: in its worked example, 12 shoppers per version detect a real 8-second gain in under 10% of simulated runs, and 400 per version in over 90%. Lesson 11, Dependence, and what it costs, shows why the sample size you report counts independent units: 60 flats in 12 buildings, treated as 60 independent flats, give an interval that catches the truth far less often than it promises. The pathway lessons of Studio 5 name the design your route declares.

Example, built from the worked example in Lesson 12: "In a randomized experiment, 800 online shoppers will be assigned by coin flip to check out with or without a one-tap reorder button, 400 per version, a size a simulation diagnosis shows detects an 8-second gain in over 90% of runs. Seconds from cart to payment will be logged for every completed checkout, and the effect will be estimated as the difference in group means, reported with its 95% interval."

4. **Your findings and what they mean.** Name precisely the primary result your declared analysis will report: the quantity, for which units, and the uncertainty it will carry. A vague promise to discuss the results later does not count. You can present expected results: what you expect that result to show, labeled as an expectation ("I expect", never "I find"), with what the expectation rests on (your hypothesis, the literature, or your EDA). A number you have not computed may appear only as an expectation with its source, never as a result. Then the broader impact and usefulness the result can have, the study's main limitations, and its future directions.

Never word an expectation as a finding: the EDA facts in part 2 are the only computed numbers this draft carries. Use no causal verb your design cannot support, and claim no population your data never reach. The words are yours: the Expo's AI guidance does not allow AI to generate your abstract, though AI may help you brainstorm and check grammar and clarity, with a ledger row for each use.

Put the title and the draft inside your milestone document, each with its character count. The review of this milestone returns them with numbered requests where they need them, and your first mentor meeting discusses them; at the next milestone you revise both for those requests, for that meeting, and for your verified result, and they go out with your application.

Milestone 6: Your data and measurement, governed

Open the milestone workbook in Colab

This chapter closes Studio 6: Govern data and measurement. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. Provenance documentation, a data-management record, your measurement specification, and a route-specific permission recheck.

What you bring

You also carry forward the last milestone's artifact: **Milestone 5: Your pathway, declared**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 20: Data Provenance and Data Quality: your acquisition route, written and dated, your provenance record for every dataset and borrowed number (the primary source behind your headline claim opened and read), your data-management update, and the permission recheck against the data as they actually arrived.
- Lesson 21: Measurement and Operationalization: your measurement specification: concept, construct, indicator, the meaning sentence for each number, the reliability check's result, the indicator-to-use validity argument with its strongest rival reading, and the boundary that follows.

The practice

1. Settle your acquisition route first, using the guidance on the studio opener, and write down which route you took and why.
2. Document provenance for every source: who produced it, how, when, and under what terms.
3. Write your measurement specification: the construct, the operationalization, and the gap between them.
4. Assess reliability and validity for your own instrument, on the evidence you actually have.
5. Recheck permissions against the data as they actually arrived, not as planned.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** Data that arrived differently from the plan can carry permissions the plan did not cover.
- **Evidence, provenance, and reproducibility.** Provenance is the evidence rail applied to your own data.
- **AI activity, verification, and human decisions.** An assistant can describe a dataset it has never seen; verify every claim against the file.
- **Uncertainty, claim boundary, and revision history.** Measurement error is uncertainty; state it alongside sampling uncertainty, not instead of it.

Where this milestone sits

You arrived carrying Milestone 5: Your pathway, declared. Milestone 7 runs the declared analysis on these governed data and produces your first reproducible result. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Any measurement change is a Contract version, because it changes what your estimand refers to.

The book's own checks

Your course rubric on Brightspace scores these checks inside its rows, so use this table to test your work before you hand it in. The points below are the book's, not your course score.

Each row scores **0, 1, or 2. 10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: Provenance documentation, a data-management record, your measurement specification, and a route-specific permission recheck	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** **Runtime** → **Run all**, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

File → **Print** opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.